perfectly secure communication, with the possibility of undetected eavesdropping forbidden by the fundamental laws of physics themselves.

Suddenly, entanglement and Bell's theorem weren't just concerns for a handful of physicists and philosophers in an abstruse and neglected corner of science. Practical questions of computing technology and cryptography were at stake, and naturally, governments and militaries took a fierce interest in the subject. Mastering control of entanglement, decoherence, and other phenomena first described by researchers in quantum foundations was potentially big business—and the race to build a quantum computer was on. The funding floodgates opened. Within a decade of Shor's breakthrough, the Department of Defense was funding a $20 million initiative in quantum information. By 2016, multiple US government agencies, both military and civilian, were funding quantum information technology; the EU was funding €1 billion of research and development in the subject; and China was testing a quantum communication satellite. Private corporations, like Google and Microsoft, were also doing work in this field. In short, quantum information processing was no longer a part of quantum foundations—it had spun off and become its own billion-dollar industry.

But little of this money was going to quantum foundations. The flood of new grants were almost entirely for developing practical things like building quantum computers themselves, not for new approaches to the measurement problem. Quantum foundations had borne this new fruit, it had proven it wasn't useless, but the advances in quantum information processing didn't have any immediate bearing on the mysteries at the heart of quantum theory. And many physicists, even those working in the new fields spun off from Bell's work, still took the Copenhagen interpretation's approach to physics, as summed up by Mermin: "Shut up and calculate!"

Quantum foundations were affecting computers—but computers were also affecting quantum foundations. In 1978, three of David Bohm's colleagues at Birkbeck College in London—Chris Dewdney, Chris Philippidis, and Basil Hiley—started looking over Bohm's old pilot-wave papers from the 1950s. Hiley had been working closely with Bohm at Birkbeck for over a